BOX, HW2

Ambiguity

For some reads, there is an ambiguous letter "N". We decide to not include k-mers with such letter.

Measures on several files

Using the provided `readfq` method, we got the following results:

```
File 1 : 1 sequence(s), cumulative length : 2221315
File 2 : 1 sequence(s), cumulative length : 29903
File 3 : 1 sequence(s), cumulative length : 9181
File 4 : 7 sequence(s), cumulative length : 2553358
File 5 : 1 sequence(s), cumulative length : 29706
File 6 : 1 sequence(s), cumulative length : 213008
```

File 1 appears to be a complete bacterial genome represented by a single sequence of approximately 2.2 million base pairs. Similarly, File 4 presents a comparable length (about 2.55 million bp) but is split across 7 sequences. The other files consist of single sequences of varying smaller lengths: around 10 to 213 thousands of base pairs. It represents smaller sequences or fragments, may be it is smaller genomes like viruses (some thousands of bases).

NCBI BLAST investigation

Using the NCBI BLAST tool, we found out the matching species for each file: 1. Streptococcus pneumoniae 2. Severe acute respiratory syndrome coronavirus 2 3. Human immunodeficiency virus 1 4. Streptococcus hyointestinalis 5. Severe acute respiratory syndrome coronavirus 2 6. Streptococcus suis

Jaccard indexes

The Jaccard index indicates how similar the two sets of k-mers are for a given pair of files. Some files do not share any 20-mers. $J(1,2)$ and $J(1,6)$ are significantly low, their 20-mers sharing are due to randomness. $J(1,4)$ and

$J(4,6)$ show a higher similarity. This reflects a true relationship: these files belong to different species of the same bacterial genome (Streptococcus). It is worth noting that $J(1,6)$ is surprisingly low for bacteria of the same genus. This is a mathematical bias rather than a biological one: because File 6 is only a small fragment, its intersection with the complete genome of File 1 is naturally limited. Meanwhile, the union remains massive, which heavily pulls the Jaccard index down. Finally, $J(2,5)$ is remarkably high. This confirms our BLAST results, proving that both files contain genomes from the exact same viral species.

Jaccard indexes : $J(1,2) = 9.170324097594257e - 07$

$J(1,4) = 0.0033671891497898485$

$J(1,6) = 2.638133971251148e-05$

$J(2,5) = 0.8285967186166345$

$J(4,6) = 0.08781022348438079$

$J(1,3) = J(1,5) = J(2,3) = J(2,4) = J(2,6) = J(3,4) = J(3,5) = J(3,6) = J(4,5) = J(5,6) = 0$

About canonical k-mers

Because DNA has a double-helix structure, a sequencing read only captures one of the two strands. However, due to base complementarity (A pairs with T, and C pairs with G), knowing one strand allows us to easily deduce the other. To avoid the bias of counting a sequence and its reverse complement as two distinct entities, we compute the reverse complement for each k-mer and keep only the lexicographically smaller one.

Abundancy histogram

The histogram shows that almost all 20-mers appear only once in the sequence (notice that the scale is logarithmic). This indicates that most k-mers are unique within the genome. As abundance increases, the number of distinct k-mers drops drastically: only a very small number of sequences occur multiple times, which corresponds to repeated elements in the DNA sequence.

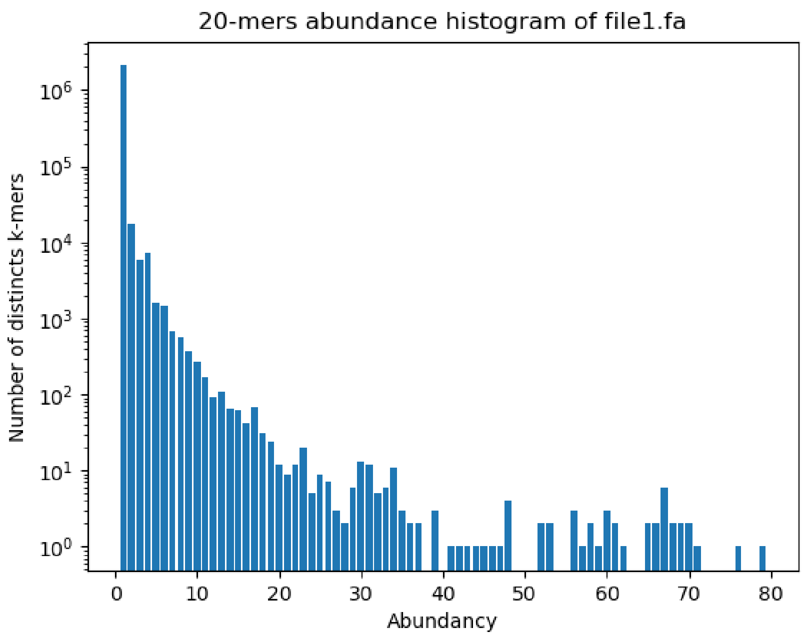

Figure 1: 20-mers abundance histogram of file file1.fa